DSCI-560: Data Science Practicum Laboratory Assignment 8
Representing Document Concepts with Embeddings
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1) Embeddings for Documents

Return to the data and the results of the previous lab on data extraction, analysis, and clustering of Reddit postings. If you used doc2vec to represent your data into vectors, these vectors are called embeddings.

Doc2vec allows the user to configure vector dimensions and values, including vector size, minimum count, and epochs (https://radimrehurek.com/gensim/models/doc2vec.html). For this task, use all the data collected from the previous lab to generate their embeddings/vectors using THREE different doc2vec configurations, including different vector sizes. Then cluster the data using the vectors and the cosine distance metric.

Once the clusters are formed, examine the data in each cluster to argue for which embedding configuration is best. There is no right or wrong answer to your question because of the subjective nature of this evaluation process. However, you are expected to do your utmost to assess and draw conclusions based on your qualitative and/or quantitative results.

2) Embeddings based on Word2Vec and Bag-of-Words

There are other ways to vectorize documents. One notable way to generate an embedding for a document is to make a vector of word frequencies. To convert a body of words into a variable-sized vector of word frequencies, you will need to group/cluster the words into a specific number of bins first. For this task, one can manually bin all words in their dataset based on the similarities and differences in their definitions. Alternatively, you may accelerate this task by using another program called Word2Vec to vectorize all words in your data set and cluster them into specific bins based on the vector distances.

Therefore, go through the tutorial on the following link to get familiar with Word2Vec and its usage.

https://www.geeksforgeeks.org/python-word-embedding-using-word2vec/

Vectorize, then cluster the words (all or selected) found in the same Reddit data used above. The dimension may be adjusted to optimize the representation's accuracy, but keep in mind that it is only used for binning/clustering.

Once the words are binned based on their distance metric, you can use them to create a vector for each document based on the normalized frequency of the words in each bin.

In other words, to generate a vector of N-dimension for a single Reddit post, you can use Word2Vec to cluster all words in all your data into K-bins. Then, for each post, you can partition all its words into K bins and create a vector with K entries, each representing the number of words in each bin. To normalize the vector, you can divide all numbers by the number of words in the post.

Using this method, embed all the data using the same 3-dimensional vectors you experimented with in Doc2Vec. Compare and contrast the quality of the embedding methodology. Which of the two methods is superior for a specific dimension?

3) **Comparative Analysis**
   Critically compare and contrast the two methods mentioned above.
   Choose a couple of ways for evaluation and justify why you are using these methods.
   Determine which one is better in representing the meanings of the documents.
   What are the advantages and disadvantages of each of these embedding methods?

4) **Team Discussions**
   Your team is expected to meet in person / virtually each day of the week and discuss the assignment progress & next steps. Document and compile minutes of all meetings in a separate file called **'meeting_notes_L8_<team_name>.pdf'**

5) **Submission**
   Make one submission per team. Each team must submit all the code files for the working solution, a readme document containing information for running the code in PDF format, and a document that outlines the minutes of all team meetings in PDF format.

   Provide a video per team that demonstrates the entire working solution and explains how the data tables were loaded, demonstrates query results, and talks about the design decisions made along with reasoning for the same. Also, include details about how your team preprocessed the data. Please include the team's name and the names of all members in the video.

   There will be a 50% penalty for all late submissions.